

检测报告

报告号: AJFS1706003401FF_CN

日期: 2017 年 6 月 15 日 第 1 页 共 8 页

北京天山新材料技术有限公司

北京市石景山区八大处高科技园区双园路 5 号

以下测试样品为客户提供并确认:

产品描述: 改性硅烷密封胶**SGS 参考号:** TP17-004214**规格:** STP1921**测试要求:**

TB/T 3237-2010 《动车组用内装材料阻燃技术条件》

1. GB/T 8323.2-2008 《塑料—烟生成—第 2 部分:单室法测量烟密度试验方法》

2. GB/T 2406.2-2009 《塑料—用氧指数法测定燃烧行为 第 2 部分: 室温试验》

3. UIC 564-2:1991 附录 4: 测定刚性非热塑性材料防火性能的测试方法

4. TB/T 3237-2010 《动车组用内装材料阻燃技术条件》条款 4.4 毒性气体分析

测试结果 -- 参见数据附表 --**测试周期:**

样品收到日期 : 2017 年 6 月 02 日

测试执行周期 : 2017 年 6 月 02 日 至 2017 年 6 月 12 日

通标标准技术服务有限公司安吉分公司

授权代表签署

邹石龙
技术经理SGS-CSTC (Shanghai) Inspection & Testing Services Co., Ltd.
Anji Branch Harbin

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检测报告

报告号: AJFS1706003401FF_CN

日期: 2017 年 6 月 15 日 第 2 页 共 8 页

1. 烟密度测试 (GB/T 8323.2-2008)

I. 测试方法

测试方法按照 TB/T 3237-2010 条款 4.3 以及 GB/T 8323.2-2008《塑料—烟生成—第 2 部分:单室法测量烟密度试验方法》进行。

II. 样品信息

样品结构	两片金属板基材中间夹有一层改性硅烷密封胶
颜色	灰色 (密封胶)
厚度	大约 3.5mm (基材+密封胶)
样品尺寸	75mm×75mm
基材	金属板, 大约 1.0mm, 银色 2PCS

预处理	温度 (°C)	相对湿度 (%)	周期 (h)
样品	23±2	50±10	>48*
背板	23±2	50±10	>48

* 直到样品的重量按标准要求达到质量恒定状态。

III. 测试结果

1) 测试模式: 辐射强度 50 kW/m², 无引燃火焰

测试样品	1 st	2 nd	3 rd	均值
D _{s1.5}	0	0	0	0
D _{s4.0}	0.1	0.2	0.2	0.1
D _{s,max}	9.2	10.6	16.8	17.6
T(D _{s,max}) 秒	1200	1200	1200	1200
D _c	2.5	1.0	0.5	0.7
D _{s10}	2.1	3.1	4.8	5.1
观察	烟气的颜色: 灰色			

2) 测试模式: 辐射强度 50 kW/m², 引燃火焰

测试样品	1 st	2 nd	3 rd	均值
D _{s1.5}	0	0	0	0
D _{s4.0}	0.2	0	0.2	0.1
D _{s,max}	15.6	20.3	16.8	17.6
T(D _{s,max}) 秒	1200	1200	1200	1200
D _c	0.9	0.8	0.5	0.7
D _{s10}	4.6	5.9	4.8	5.1
观察	烟气的颜色: 灰色			

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备注:

$D_{s1.5}$ —测试开始 1.5min 时的比光密度;

$D_{s4.0}$ —测试开始 4.0min 时的比光密度;

$D_{s,max}$ —整个测试期间 (20min) 内最大比光密度;

$T(D_{s,max})$ —发生最大比光密度 $D_{s,max}$ 时的时间, 单位秒;

D_c —清晰光束校正因子;

D_{s10} —测试开始 10min 时的比光密度。

内装材料燃烧后烟密度指标应满足下表要求:

燃烧方式	$D_{s1.5}$	$D_{s4.0}$
无引燃火焰	≤ 100	≤ 200
引燃火焰	≤ 100	≤ 200

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检测报告

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日期: 2017 年 6 月 15 日 第 4 页 共 8 页

2. 氧指数测试 (GB/T 2406.2-2009)

I. 测试方法

本试验方法按照 GB/T 2406.2-2009 《塑料 用氧指数法测定燃烧行为 第 2 部分: 室温试验》进行。

II. 试件描述

样品结构 / 颜色	两片金属板基材中间夹有一层改性硅烷密封胶 / 银色		
试样类型	III		
试样尺寸	长度: 100mm	宽度: 10mm	厚度: 2.8mm
	数量: 15 个		

	温度	湿度	时间
预处理	23±2 °C	50±5% R.H.	>88小时
测试条件	23±2 °C	--	--
点燃方法	顶端点燃法		

III. 测试结果

氧指数 [浓度, % (V/V)]: >60 [圆整到0,1 % (V/V)]

初始氧浓度的确定 (≤1%变化幅度内)

氧气浓度, % (V/V)	30	40	50	60	61
燃烧长度 (毫米)	<50	<50	<50	<50	<50
结果 ("X" 或 "O")	O	O	O	O	O

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检测报告

报告号: AJFS1706003401FF_CN

日期: 2017 年 6 月 15 日 第 5 页 共 8 页

3. UIC 564-2:1991 燃烧性能试验

I. 样品信息

样品结构	两片金属板基材中间夹有一层改性硅烷密封胶
颜色	灰色 (胶)
厚度	3.5mm (基材+胶)
样品尺寸	160mm×400mm
基材	金属板, 大约 1.0mm, 银色 2PCS

预处理	温度 (°C)	相对湿度 (%)	周期 (h)
	20~24	50~65%	>48

II. 测试方法

该测试按照 UIC 564-2:1991 附录 4:测定钢性非热塑性材料防火性能的测试方法进行。

III. 测试结果

测试项目	纵向		横向		平均值
	1	2	3	4	
酒精火焰熄灭后, 样品续燃时间 (s)	0	0	0	0	0
样品阴燃时间 (s)	0	0	0	0	0
火焰烧毁表面区域面积 (cm ²)	0	0	0	0	0
燃烧颗粒或滴落物落下 (是/否)	否	否	否	否	-
样品是否被烧穿至上表面 (是/否)	否	否	否	否	-

评级: A 级

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检测报告

报告号: AJFS1706003401FF_CN

日期: 2017 年 6 月 15 日 第 6 页 共 8 页

材料测试按照下述的要求进行分类

持续燃烧时间 (s) (1) 表面区域 (cm ²) (2)	P≤2	2<P≤10	P>10
S≤100			
100<S≤150			
S>150			

(1)P=酒精火焰熄灭后的样品续燃时间, 以秒计;

(2)S=测试后样品表面的损毁面积, 以 cm² 计。

A 级:

所有测试样品各自结果符合白框内要求, 且无样品出现下列现象:

- 烧穿至样品上部表面;
- 观察到燃烧的碎片或滴落物;
- 观察酒精火焰熄灭后, 出现时间超过 10 秒的阴燃。

B 级:

所有样品的续燃时间平均值和损毁区域面积平均值符合灰色框或白色框内要求, 且无样品出现下列现象:

- 烧穿至样品上部表面;
- 记录有释放燃烧的碎片或滴落物;
- 观察酒精火焰熄灭后, 出现时间超过 10 秒阴燃。

C 级:

所有测试结果不满足 A 级和 B 级的情况, 都归类为 C 级。

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报告号: AJFS1706003401FF_CN

日期: 2017 年 6 月 15 日 第 7 页 共 8 页

4. TB/T 3237-2010 毒性气体分析

I. 测试方法

测试方法按照 TB/T 3237-2010《动车组用内装材料阻燃技术条件》条款 4.4—毒性气体分析所述要求进行。

II. 样品信息

样品结构	两片金属板基材中间夹有一层改性硅烷密封胶
颜色	灰色（密封胶）
厚度	大约 3.5mm（基材+密封胶）
样品尺寸	大约 75mm×75mm
基材	金属板，大约 1.0mm，银色 2PCS

预处理	温度 (°C)	相对湿度 (%)	周期 (h)
样品	23±2	50±10	>48*
背板	23±2	50±10	>48

* 直到样品的重量按标准要求达到质量恒定状态。

III. 测试结果

a) 测试面: 任意面

b) 测试模式: 辐射照度为 25 kW/m², 有引燃火焰

气体种类	要求浓度 mg/m ³ (ppm)	实测值 (ppm)	结果
CO	<4000 (3500)	10	符合
CO ₂	<90000 (50000)	3100	符合
HF	<82 (100)	ND	符合
HBr	<330 (100)	ND	符合
HCl	<150 (100)	ND	符合
NO _x (以NO ₂ 计)	<190 (100)	1	符合
SO ₂	<260 (100)	ND	符合
HCN	<110 (100)	ND	符合

注: ND=未检测到。

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日期: 2017 年 6 月 15 日 第 8 页 共 8 页

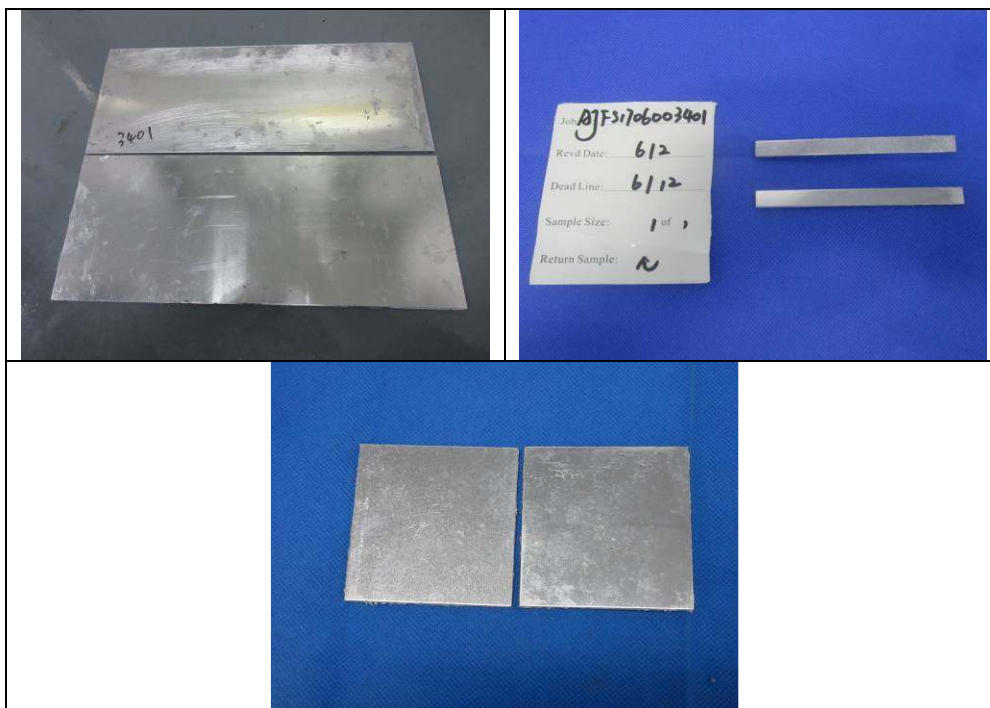
陈述:

本试验结果得出的产品性能是在特殊条件下的检验结果。在实际应用中, 它们不能单独作为评价该产品潜在火灾、烟气和毒性危险性的依据。

本测试结果仅代表被测试试样。成分、厚度等试样的任何微小的变化可能会导致测试结果的巨大变化, 因此此测试结果对此无效。实际应用时, 材料的提供和使用上需要格外小心以保证使用的材料充分代表测试试样。

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